

# Real-Time Gaze Awareness in Conversational Agents: Enhancing Collaboration and Personalization in VR Art Experiences

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**Abstract.** The increasing integration of conversational agents into interactive environments has prompted the investigation of methods to enhance user experience. This study explores how varying levels of real-time gaze awareness influence interactions with LLM-based conversational agents within a Virtual Reality (VR) art exhibition. Fifty-one participants were randomly assigned to one of three experimental groups, each experiencing a VR exhibition featuring five paintings and interacting with a conversational agent. The groups differed based on the agent's gaze-awareness capabilities: 1) identifying the painting the user was viewing, 2) responding based on specific areas within a painting, and 3) analyzing detailed gaze patterns, including fixation counts, area transitions, and dwell times. Results revealed that participants in the third group, where detailed gaze pattern analysis was employed, perceived their interactions as significantly more personalized and collaborative. However, this did not lead to measurable differences in reported enjoyment, engagement, or perceived gaze awareness compared to the other conditions.

**Keywords.** Eye Tracking, Conversational Agents, Large Language Models, Human-Agent Interaction, Virtual Reality, Cultural Heritage

## 1. Introduction

Virtual Reality (VR) opens new avenues for immersive and interactive art experiences, enabling museums to engage visitors in deeper and more meaningful ways [1–3]. However, a fundamental challenge remains: *How can these systems identify what captures a visitor's attention and adapt the experience accordingly?*

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Eye-tracking technology offers a powerful way to address this challenge. By analyzing where visitors focus their gaze, it provides valuable insights into engagement and cognitive responses [4–7]. Some museums leverage gaze data for post-visit feedback, yet many implementations remain passive, missing opportunities to dynamically enhance real-time interactions [8]. By integrating eye-tracking with AI-driven virtual agents, museums could enable context-sensitive interactions, such as delivering personalized commentary or supplementary information based on a visitor's gaze [8].

While prior research has demonstrated the benefits of eye-tracking in virtual environments [9, 10], its integration with conversational agents—digital guides capable of real-time dialogue has not been fully explored. Virtual agents are already used to provide information and facilitate engagement [11, 12], but combining real-time gaze data with conversational agents could further enhance the user experience by enabling more adaptive and responsive interactions.

This study examines this potential in a VR art exhibition featuring five paintings, addressing the main question: *At what level of gaze awareness can conversational agents improve the user experience?* Fifty-one participants were divided into three experimental conditions: (1) basic gaze awareness, where the agent detects which painting the user is viewing; (2) an enhanced model recognizing specific areas of interest (AOIs) within paintings; and (3) an advanced system analyzing gaze patterns, including dwell time, transitions between AOIs, and fixation counts. Following the VR session, participants completed questionnaires assessing engagement, enjoyment, perceived personalization, and collaboration. These quantitative measures were supplemented with structured interviews for qualitative insights.

The study seeks to answer the following research questions: (1) *How can a conversational agent be designed to integrate real-time eye-gaze data for adaptive and personalized interactions in a VR art exhibition?* (2) *To what extent do different levels of gaze-awareness influence user enjoyment, engagement, perceived personalization, and sense of collaboration?* By exploring these questions, this work contributes to the design of gaze-aware hybrid intelligent systems that promote shared agency, responsiveness, and personalization in immersive cultural heritage environments.

## 2. Related Work

*Eye Tracking and Its Applications* Eye tracking is a technique used to detect and measure eye movements, providing insights into visual attention and associated behaviors [13]. An eye tracker identifies a user's gaze point and records the corresponding coordinates. This technology has found applications across diverse fields, including psychology, where it helps analyze cognitive and emotional processes [7, 14], neuroscience for studying neural responses to visual stimuli [15], education for understanding learning patterns [16–18], and human-computer interaction (HCI) for enhancing user interfaces and interactions [8, 19, 20]. In cultural heritage studies, eye tracking has been utilized to examine visitor behaviors in museums, assess cognitive engagement with artworks, and analyze learning processes [21–26]. Additionally, it has supported the development of personalized experiences by modeling visual attention, identifying areas of interest (AOIs), and creating targeted recommendations [27–30]. Beyond these applications, eye tracking has been used for emotion recognition and personality detection [31–34], as

well as studying attentional mechanisms, cognitive processing, decision-making, and memory formation [7, 35–42]. Recently, the integration of eye tracking with virtual reality (VR) and mobile platforms has opened new avenues for research. These include immersive environments for training, personalized content delivery, and advanced user behavior analysis [9, 22, 43–49].

*Multimodal Interaction with Eye Tracking* Multimodal interaction enables users to engage with systems through various input modes, such as vision, speech, and touch. It has been shown to enhance user experience, offer flexibility in interaction methods, and adapt to diverse preferences [50]. Among these modalities, the combination of vision and hearing has proven particularly effective in HCI systems [26]. Eye tracking has been a valuable tool in multimodal systems, particularly in cultural heritage applications. For example, it has been employed in museum guides to detect when visitors view specific objects and deliver tailored information accordingly [29, 51]. Studies have combined mobile eye tracking with image-based object recognition to track visitor gaze on exhibits, identify their locations, and provide verbal content relevant to the exhibit of interest [29, 30]. Similarly, Pierdicca et al. [28] utilized gaze data to develop visitor attention models, extract behavioral patterns, and recommend artwork. Raptis et al. [26] extended these applications by allowing interaction with AOIs within exhibits, enabling visitors to request detailed cultural information and revisit key points of interest.

*Research Gap and Contribution* While prior work has integrated eye tracking into multimodal systems, no study, to the best of our knowledge, has explored the combination of large language models (LLMs) with gaze data for enhancing user interaction in VR art exhibitions. LLMs have been utilized in virtual environments to facilitate natural language interactions, offering context-aware responses and the ability to process complex inputs [52–55]. These capabilities make LLMs an ideal complement to gaze-aware systems, which can capture user attention in real time. This study addresses the gap by integrating LLMs with gaze-tracking technologies to design gaze-aware multimodal interactions within VR art exhibitions. Specifically, it investigates how varying levels of gaze awareness impact user experiences, contributing to the development of hybrid intelligent systems [56] that support adaptive and collaborative interactions. By bridging gaze data and advanced natural language processing, this research aims to offer an approach to enhancing user experience in immersive cultural heritage environments.

### 3. Methodology

This section describes the methodology employed to address the research questions. It begins with an overview of the platform used for the user study, followed by a detailed explanation of the experimental setup, procedures, and evaluation measures applied to assess the system.

#### 3.1. System Overview

Figure 1 illustrates an overview of the integrated system, which combines multiple technologies to deliver an immersive user experience. At its core lies a virtual exhibition,

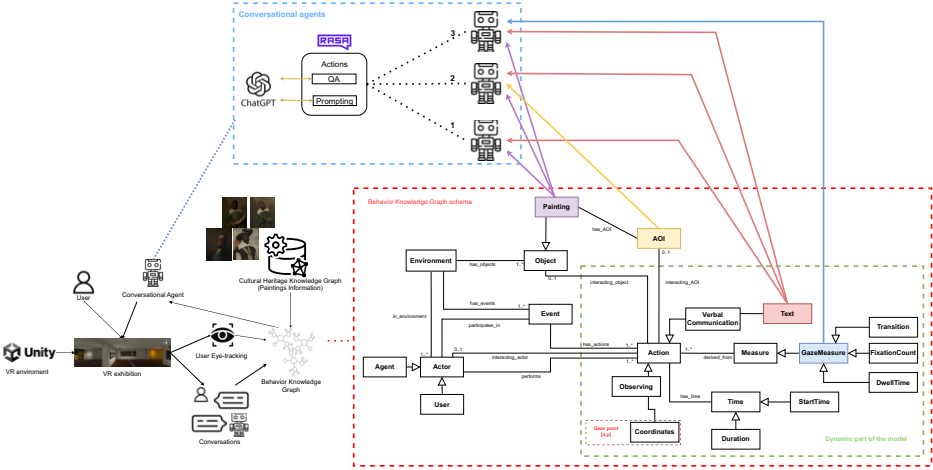


Figure 1. System Overview and Knowledge Graph Schema

developed using Unity<sup>2</sup>, where users can explore 3D models of paintings and access contextual information through a conversational AI-powered virtual guide. This guide leverages a dynamic knowledge graph that links metadata—such as descriptions, historical context, and Areas of Interest (AOIs) within the paintings—to the user’s gaze behavior, tracked using eye-tracking technology.

3.1.1. Knowledge Graphs

**Artwork Metadata** Each painting in the virtual exhibition is enriched with metadata stored in a knowledge graph. This includes the title, description, and narrative, along with predefined Areas of Interest (AOIs)—notable regions like people or objects that attract attention. The exhibition features five curated paintings from “*HERE: Black in Rembrandt’s Time*” [57], modeled as individual 3D objects with textual descriptions.

**Behavior Metadata** Figure 1 depicts a conceptual model of the VR environment, designed to capture both the static structure and the dynamic interactions of users and agents. Key concepts include Actor, Environment, Object, Event, Action, and Measure, chosen for their importance in modeling the spatial and behavioral dimensions of VR experiences. The Environment represents the overarching static structure, containing Objects (e.g. paintings) that may have specific Areas of Interest (AOIs). Within this environment, an Actor (either a human user or a virtual agent) participates in Events and performs Actions. Dynamic interactions are modeled through Actions, which are associated with time (StartTime and Duration) and may include Verbal communication with another Actor or Observation of an Object or AOI. Observations are further detailed by Coordinates, representing gaze points, which enable the derivation of various Measures, such as FixationCount, DwellTime, and Transition. These measures provide quantitative insights into user behavior and interaction patterns. The relationships between these concepts reflect how users engage with the VR environment. For example, an Actor participating in an Event might interact

<sup>2</sup><https://unity.com/>

with objects, producing measures of attention and focus. This integration of static and dynamic components offers a comprehensive framework for analyzing VR experiences.

### 3.1.2. Real-Time Eye-Tracking Data

Users wear an HTC VIVE Pro Eye headset equipped with an integrated eye-tracking system. Using the Tobii Pro SDK,<sup>3</sup> the system captures real-time gaze data pre frame. By employing raycasting techniques, the SDK precisely tracks users' gaze by detecting intersections with virtual objects, capturing collision points and corresponding texture coordinates. This data collection process includes parameters such as headset position, texture coordinates, precise timestamps, and object identifiers (e.g., "C6.Painting"). Gaze data are recorded as 3D coordinates  $[x_i, y_i, z_i]$ , representing the user's view in the virtual space. When the gaze intersects with a painting, the gaze data is projected onto the painting's surface, converting it into 2D coordinates  $[x_j, y_j]$ , allowing for a generalized and environment-independent representation. To ensure comparability across paintings of varying sizes, gaze coordinates are normalized within a range of  $[0.00, 0.00]$  (top-left corner) to  $[1.00, 1.00]$  (bottom-right corner). In addition, temporal data, including the start time and duration of each observation, is also captured, enabling analysis of gaze patterns, such as the sequence of viewed objects and areas of sustained attention. For further analysis, only gaze points with a duration exceeding 100 milliseconds were considered, following established practices [58, 59].

### 3.1.3. Gaze-Aware Conversational Agent for Adaptive User Interaction

Three conversational agents were implemented to investigate the effects of different levels of gaze awareness, leveraging OpenAI GPT-4o mini<sup>4</sup> and the RASA<sup>5</sup> framework. These agents function as virtual museum guides, tasked with providing engaging and informative interactions to enhance user experiences in a virtual reality (VR) exhibition. The agents are programmed to perform two primary actions: answering user questions and initiating conversations. Each action employs tailored prompts specific to the capabilities and roles of the agents. The agents are designed to differ in their utilization of gaze data, which informs their responses and interaction strategies. Below, we provide an overview of their shared components and platform-specific implementations. Due to space constraints, only the prompt for initiating conversations is detailed.

**Shared Components** The shared prompt components define the general role (e.g., "*You are an AI assistant serving as a virtual museum guide in a VR exhibition featuring five unique paintings. Your task is to engage users by encouraging interaction with the artworks and providing insightful information to enhance their experience.*") and behavior of the agents across all platforms. These components include instructions for interacting with users and guidelines for maintaining engaging and informative dialogue. They also encompass *conversation histories*, stored turn-by-turn in the *knowledge graph* (Section 3.1.1), enabling the system to reference prior user interactions and adapt responses accordingly. To ensure brevity, the conversation history is limited to the past 20 turns, encompassing exchanges between both the agent and the user.

<sup>3</sup><https://developer.tobiipro.com/>

<sup>4</sup><https://openai.com/>

<sup>5</sup><https://rasa.com/>

**Platform-Specific Details** In the first platform implementation, the agent uses basic gaze awareness—determined by tracking the user’s gaze point on a painting—to identify which artwork the user is observing and prioritize providing information about it (i.e., “The user is currently observing a painting with the following details: (GRAPH.get\_last\_obj(actorID)). Start by providing information about the painting the user is observing.”). The agent retrieves details about the user’s last observation (gaze point on the painting) from the knowledge graph, enabling it to tailor its responses.

The second platform implementation, in addition to paintings, focuses on specific areas of a painting that capture the user’s attention and prioritizes those areas when providing information (i.e., “They are specifically focused on this area of the painting: (GRAPH.get\_last\_aoi(actorID)). Start by providing information about the specific area of the painting the user is observing.”). It extracts information about the most recent area the user is looking at (gaze on AOI) from the knowledge graph, along with details about the painting itself.

The third platform implementation uses gaze pattern data to tailor responses based on user engagement levels and patterns. The prompt incorporates detailed metrics, including time spent on AOIs, transitions between them, and fixation counts. These metrics are calculated and extracted from the knowledge graph in real-time, guiding the agent to adapt responses that align with the user’s interaction style. For example, if a user demonstrates high engagement with a specific AOI (evidenced by numerous fixations and minimal transitions), the agent provides detailed insights about that area, including its context and significance. Conversely, if the user shows low engagement, characterized by brief interactions and frequent transitions, the system encourages further exploration by suggesting alternate AOIs or probing for interests. The prioritization strategy also addresses scenarios such as quickly scanning, comparing multiple areas, or spending prolonged periods in unfamiliar areas:

```
The user is currently observing a painting with the following details:({GRAPH.get_last_obj(
    actorID)}). Leverage the following data to tailor your response:
- Time spent on various areas of the painting:
{GRAPH.get_time_spent_on_aois(last_obj_name, actorID)}
- Number of transitions between different areas:
{GRAPH.get_number_of_transition_between_AOIs(last_obj_name, actorID)}
- Number of fixations on specific areas:
{GRAPH.get_number_of_fixation_count(last_obj_name, actorID)}

### Prioritization:
- Start with the followings:
1. Highly Engaged with a Specific Area: - If the user has many fixations on one area
    with few transitions between areas, provide detailed insights about this Area of
    Interest (AOI), including its context, significance, or interesting facts.
2. Scanning Quickly: - If the user spends little time on areas but makes frequent
    transitions between them, offer a broad overview, highlighting major themes or
    features for an accessible introduction.
3. Comparing Different Areas: - If the user has frequent fixations on two or more areas
    and transitions between these areas, describe connections or contrasts of these areas
    to support their exploration.
4. Unfamiliar with a Specific Area: - If the user has few fixations but spends a
    relatively long time in one area, provide introductory information to guide their
    understanding of that AOI
5. Low Engagement: - If the user shows minimal engagement (few fixations, short time
    spent, and frequent transitions), encourage further interaction by suggesting another
    AOI or asking about their interests.
- Once all relevant details about the current painting and its areas have been shared, guide
    the conversation toward exploring other topics or artworks.
```

The action to initiate interaction is triggered every 60 seconds in the VR environment and validated in RASA. If no interaction occurs between the agent and the user within the preceding 30 seconds, based on data extracted from the knowledge graph, the action

is executed. The question-answering prompt functions similarly, except the agent's role is to answer questions, prioritizing the user's input.

### 3.2. Experiment Setup

**VR Visit** The study, approved by the university's ethics committee, ensured anonymity and confidentiality through written consent. Participants were university students and employees, each of whom received a 10-euro voucher as compensation for their participation. They explored the VR exhibition at their own pace, viewing paintings and reading texts, while being assigned to one of three platforms based on their order of arrival. Participants could ask questions, follow the virtual agent's prompts, or simply listen to its interactions. Voice and eye-gaze data were recorded. After the visit, participants completed a questionnaire and provided feedback in interviews.

**Questionnaire** The questionnaire began with demographic questions (age, gender) and assessed participants' familiarity with VR, museum visit frequency, and knowledge of the paintings. It included behavioral questions on interaction preferences with guides (active engagement vs. passive listening) and evaluated the agent's effectiveness using 5-point Likert scale statements (from Strongly Disagreed to Strongly Agreed).

**Interview** Participants were first asked about their overall impressions of the VR experience and interactions with the virtual agent. They were then questioned on whether they noticed the agent's gaze-tracking and contextual interpretation. The discussion covered what participants recalled, whether they learned new information, and if the agent's content was useful. Finally, they were asked if the experience felt personalized to their interests.

### 3.3. Measures

We evaluate the virtual agents based on five key measures: enjoyment, engagement, perceived personalization, sense of collaboration, and gaze awareness. Table 1 lists the questions used for each measure. Additionally, we analyze the time spent in the environment, interaction count, and interview data to gain a comprehensive understanding of user experiences.

**Enjoyment** measures user satisfaction and pleasure from interacting with the virtual agent, influencing their likelihood of returning. It is assessed through questions on general pleasure ( $Q_1$ ), future interaction preference ( $Q_2$ ), curiosity about the exhibition ( $Q_3$ ), educational value ( $Q_4$ ), and overall satisfaction ( $Q_5$ ). **Engagement** evaluates how involved and absorbed users feel during their interaction, reflecting emotional and cognitive investment. We use questions that address time distortion ( $Q_1$ ), agent attentiveness ( $Q_2$ ), and interest in the agent's responses ( $Q_3$ ) [60]. **Perceived personalization** evaluates how much the user feels the virtual agent adapts to their preferences and needs. It is assessed through questions on the agent's adaptability ( $Q_1$ ) and awareness of the user's interests ( $Q_2$ ).

**Collaboration** is a process in which multiple actors work together to achieve shared goals. In hybrid intelligent systems, it includes components such as *Shared effort*, the sense of working together ( $Q_1$ ); *Mutual understanding*, which reflects the system's understanding of the user's goals ( $Q_2$ ) and the user's understanding of the system's capa-

bilities ( $Q_3$ ); and *Responsiveness*, the system's ability to adjust based on user input ( $Q_4$ ). **Gaze awareness** measures the user's perception that the virtual agent is aware of their visual attention during the interaction. It evaluates whether the agent can detect and respond to the user's focus, contributing to presence and interactivity. Questions assess the agent's understanding of the user's focus ( $Q_1$ ) and its ability to respond to shifts in attention ( $Q_2$ ). **Time spent in the environment** reflects user engagement, while **number of interactions** tracks the frequency and depth of exchanges, indicating engagement and system usability. **Interview data** captures user reflections, preferences, and suggestions, offering deeper insights into perceived personalization, gaze awareness, and areas for improvement.

#### 4. Questionnaire Results

This section summarizes the findings from 51 participants. The age distribution was: 1 participant aged 18–19, 36 aged 20–30, 12 aged 30–40, and 3 aged 40–50. The sample included 28 men, 22 women, and 1 participant who preferred not to disclose their gender. Regarding VR familiarity, 2 participants were unfamiliar with VR, 10 had heard of it, 26 had tried it, and 13 were very familiar. Concerning museum visitation habits, 1 participant never visited museums, 16 visited rarely, 31 occasionally, and 3 frequently. Of the paintings presented in the VR experience, 36 were unfamiliar, 10 were vaguely familiar, 4 were somewhat familiar, and 1 was very familiar. Interest in virtual agents was expressed by 23 participants, while 4 were not interested and 24 were uncertain. For personalized virtual agents, 38 participants expressed interest, 4 were not interested, and 9 were unsure.

The results of the Likert-scale evaluation are summarized in Table 1 and visualized in Figure 2, which presents boxplots showing the distribution and mean values for each group. To compare scores across platforms and assess potential differences, a one-way Analysis of Variance (ANOVA) [61] was performed. Following significant ANOVA results, Tukey's Honest Significant Difference (HSD) [62] test was conducted for post-hoc analysis. ANOVA revealed a significant difference in **enjoyment**, with Tukey's HSD indicating that Group 2 enjoyed the experience significantly more than Group 1. No significant differences were found between Groups 1 and 3 or Groups 2 and 3. For **engagement**, no significant differences were observed; hence, no post-hoc tests were performed, suggesting similar engagement levels across groups. For **perceived personalization**, ANOVA showed significant differences. Tukey's HSD identified significant differences between Group 1 and Group 3 and between Group 2 and Group 3, indicating that Group 3 was rated as significantly more personalized. Regarding **sense of collaboration**, ANOVA revealed significant differences, with Tukey's HSD showing that Group 3 scored significantly higher than Group 1, while no significant differences were found between the other groups. For **gaze awareness**, no significant group differences were found, with mean scores of 3.21, 3.29, and 3.38 for Groups 1, 2, and 3, respectively, indicating consistent perceptions across groups. The average **time spent in VR** was 633.78, 699.19, and 729.35 seconds for Groups 1, 2, and 3, respectively. Although the mean time increased across groups, ANOVA revealed no significant differences ( $F = 0.39$ ,  $p = 0.679$ ). Similarly, the mean **number of interactions** (11.35, 15.53, and 14.47) showed no significant group differences ( $F = 0.05$ ,  $p = 0.949$ ).

**Table 1.** Evaluation Measures and Questions with Average Scores and ANOVA Results

| Measure                          | Related Questions                                                                                                                                         | AVG <sub>G1</sub>     | AVG <sub>G2</sub>     | AVG <sub>G3</sub>     | ANOVA                                    |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------|-----------------------|------------------------------------------|
| <b>Enjoyment</b>                 | Q <sub>1</sub> : I enjoyed interacting with the virtual agent.                                                                                            | Q <sub>1</sub> : 3.06 | Q <sub>1</sub> : 3.65 | Q <sub>1</sub> : 3.25 | F-statistic = 3.3921<br>p-value = 0.0419 |
|                                  | Q <sub>2</sub> : I would prefer future exhibitions to include such interactive agents.                                                                    | Q <sub>2</sub> : 3.29 | Q <sub>2</sub> : 3.88 | Q <sub>2</sub> : 3.18 |                                          |
|                                  | Q <sub>3</sub> : The virtual agent made me want to visit the real exhibition.                                                                             | Q <sub>3</sub> : 3.24 | Q <sub>3</sub> : 3.65 | Q <sub>3</sub> : 3.59 |                                          |
|                                  | Q <sub>4</sub> : I learned something new from the virtual agent.                                                                                          | Q <sub>4</sub> : 4.12 | Q <sub>4</sub> : 4.35 | Q <sub>4</sub> : 4.18 |                                          |
|                                  | Q <sub>5</sub> : I am satisfied with my interaction with the virtual agent.                                                                               | Q <sub>5</sub> : 3.06 | Q <sub>5</sub> : 3.59 | Q <sub>5</sub> : 3.29 |                                          |
|                                  |                                                                                                                                                           | <b>Avg: 3.35</b>      | <b>Avg: 3.82</b>      | <b>Avg: 3.52</b>      |                                          |
| <b>Engagement</b>                | Q <sub>1</sub> : During the interaction with the virtual agent, I lost track of time.                                                                     | Q <sub>1</sub> : 3.59 | Q <sub>1</sub> : 3.47 | Q <sub>1</sub> : 3.18 | F-statistic = 0.7066<br>p-value = 0.4984 |
|                                  | Q <sub>2</sub> : The virtual agent seemed interested in what I had to say.                                                                                | Q <sub>2</sub> : 2.47 | Q <sub>2</sub> : 3.12 | Q <sub>2</sub> : 3.12 |                                          |
|                                  | Q <sub>3</sub> : I was interested in hearing what the virtual agent had to say.                                                                           | Q <sub>3</sub> : 4.06 | Q <sub>3</sub> : 4.00 | Q <sub>3</sub> : 3.71 |                                          |
|                                  |                                                                                                                                                           | <b>Avg: 3.37</b>      | <b>Avg: 3.53</b>      | <b>Avg: 3.33</b>      |                                          |
| <b>Perceived Personalization</b> | Q <sub>1</sub> : I felt that the virtual agent adapted its behavior based on my reactions.                                                                | Q1: 2.65              | Q1: 2.59              | Q1: 3.48              | F-statistic = 5.8234<br>p-value = 0.0054 |
|                                  | Q <sub>2</sub> : The virtual agent seemed aware of my interests during the interaction.                                                                   | Q2: 2.29              | Q2: 2.70              | Q2: 2.94              |                                          |
|                                  |                                                                                                                                                           | <b>Avg: 2.47</b>      | <b>Avg: 2.65</b>      | <b>Avg: 3.21</b>      |                                          |
| <b>Sense of Collaboration</b>    | Q <sub>1</sub> : The agent's responses made me feel like we were working together to explore the exhibits.                                                | Q <sub>1</sub> : 2.65 | Q <sub>1</sub> : 3.29 | Q <sub>1</sub> : 3.18 | F-statistic = 5.1895<br>p-value = 0.0091 |
|                                  | Q <sub>2</sub> : The virtual agent understood what I wanted and helped me achieve my goal.                                                                | Q <sub>2</sub> : 2.65 | Q <sub>2</sub> : 3.47 | Q <sub>2</sub> : 3.35 |                                          |
|                                  | Q <sub>3</sub> : It was clear to me what the virtual agent could do.                                                                                      | Q <sub>3</sub> : 3.59 | Q <sub>3</sub> : 3.59 | Q <sub>3</sub> : 4.17 |                                          |
|                                  | Q <sub>4</sub> : The virtual agent felt like it was paying attention to me during the interaction.                                                        | Q <sub>4</sub> : 3.35 | Q <sub>4</sub> : 3.41 | Q <sub>4</sub> : 4.06 |                                          |
|                                  |                                                                                                                                                           | <b>Avg: 3.06</b>      | <b>Avg: 3.44</b>      | <b>Avg: 3.69</b>      |                                          |
|                                  |                                                                                                                                                           |                       |                       |                       |                                          |
| <b>Gaze Awareness</b>            | Q <sub>1</sub> : I felt the virtual agent understood my focus during the interaction.                                                                     | Q <sub>1</sub> : 3    | Q <sub>1</sub> : 3.06 | Q <sub>1</sub> : 3.12 | F-statistic = 0.1342<br>p-value = 0.8747 |
|                                  | Q <sub>2</sub> : The virtual agent seemed to recognize when I focused on an object for an extended period or when I shifted my attention between objects. | Q <sub>2</sub> : 3.41 | Q <sub>2</sub> : 3.53 | Q <sub>2</sub> : 3.65 |                                          |
|                                  |                                                                                                                                                           | <b>Avg: 3.21</b>      | <b>Avg: 3.29</b>      | <b>Avg: 3.38</b>      |                                          |

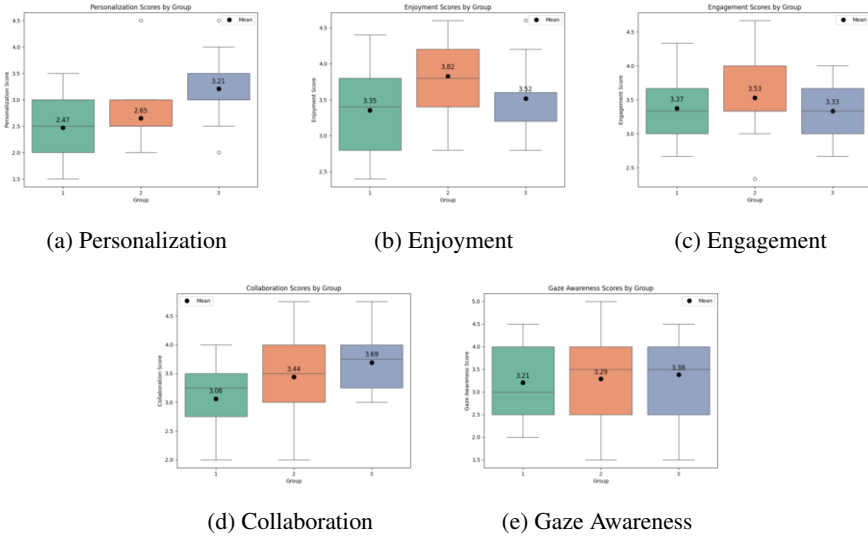
## 5. Discussion & Conclusion

The findings of this study shed light on the impact of integrating real-time eye-gaze data into interactions with conversational agents within a VR art exhibition.

### *RQ<sub>1</sub> : Design Insights*

The integration of gaze tracking into language model-driven agents presents significant opportunities for enhancing interactive experiences. However, our findings highlight key design challenges and areas for improvement.

*Issues of Initiation, Timing, and Cognitive Overload* Participants appreciated the agent's ability to provide detailed, context-rich descriptions of painting elements such as clothing, symbols, and color palettes. However, this richness was sometimes overshadowed by the agent's proactive approach, which disrupted users' focus while they were reading text or observing artwork. The agent was programmed to initiate interactions every minute if users had not engaged within the previous 30 seconds, but many found this frequency intrusive. As one participant noted, "*I honestly didn't really like that agent because I felt like I was being interrupted.*" Several participants expressed a desire for



**Figure 2.** Results- Boxplot showing distribution and means for each group

more time to process information or explore the artwork independently before receiving additional input. The simultaneous demands of navigating the VR environment, listening to the agent, and reading descriptive text often resulted in cognitive overload. One participant described the challenge: “*I didn’t know whether to focus on the agent’s speech or the text I was reading, and I ended up losing track of both.*” While the agent generally succeeded in identifying the painting a user was viewing, occasional mismatches and response delays disrupted the experience. In VR, these delays occurred because the system processed user requests via Rasa and the OpenAI API, sometimes causing the agent to reference a previous painting rather than the one currently in view. As one participant explained, “*I was confused at one point because I had already moved on, and he started talking about the previous painting.*” Future iterations should prioritize user autonomy by incorporating adaptive timing mechanisms, such as subtle, user-initiated triggers, rather than rigid, predefined intervals. This approach would allow users to engage with the agent at their own pace, reducing interruptions and cognitive overload.

**Fixation Thresholds and Improving Contextual Reasoning** The system occasionally misinterpreted brief glances or momentary attention as indicators of interest, leading to confusing responses. The gaze duration threshold for fixation was set at 100 ms, which was insufficient for determining focused attention for some participants, particularly at the beginning of interactions when they were navigating the VR environment and searching for where to go. As a result, the agent sometimes assumed interest in specific paintings or areas when users were merely glancing. On the second platform, for example, a participant began receiving information about the gilt garment in a painting despite not focusing on it:

*“The **garment** worn by Dom Miguel de Castro is richly adorned with silver gilt embroidery, reflecting his wealth and status during the 17th century. The plain falling band, made from sheer white fabric, adds to the elegance of his attire, highlight-*

*ing the refinement typical of high-ranking individuals of that era. Would you like to explore other areas of the painting or perhaps move on to another artwork in the exhibition?"*

This led to moments of confusion, where the participant was unsure which painting the agent was referencing, stating, "*Which painting were you just talking about?*" On the third platform, the agent sometimes stated, "*It seems you've spent a significant amount of time observing ...*", which participants occasionally found inaccurate. For example, when a participant focused on specific areas for the following durations (in seconds): 0.413, 0.34, 0.57, and 0.21 for the "*Red ostrich feather*," "*Cavalier hat*," "*Gilt garment*," and "*Dom Miguel*," respectively, with fixation counts of 2, 1, 3, and 1, the agent provided the following response:

*"It seems you spent a considerable amount of time focusing on the **gilt garment**, which is ornately decorated with silver gilt embroidery—showcasing wealth and elegance typical of the 17th century. The **red ostrich feather**, a symbol of status and luxury, also caught your attention, highlighting Dom Miguel de Castro's high societal standing. If you're ready, I can guide you to other fascinating artworks in this exhibition! Would you like to explore something else?"*

While the response accurately reflected the highest gaze durations, some participants did not perceive these durations as significant. In some cases, participants were reading the descriptive text beneath the painting when they received this message, which led to confusion. Incorporating contextual awareness, such as recognizing when participants are reading or browsing, and implementing a threshold for what constitutes a "significant" duration could help refine gaze interpretation and improve response accuracy.

**Limitations** Some users struggled with the interaction mechanics, such as the need to hold a button and speak slowly for the agent to respond accurately. This was frequently described as frustrating and disrupted the natural flow of interaction. Additionally, participants noted that the agent's voice sounded unnatural and robotic, which reduced immersion. Some users also expressed disappointment with the limited selection of paintings, as they expected greater variety. Another recurring issue was the agent's use of phrases like "as we discussed", which caused confusion when no prior discussion had taken place. Furthermore, the agent often repeated the same statements in its responses or occasionally provided redundant information, primarily due to the limited data available about the paintings.

## *RQ<sub>2</sub> : Impact on User Experience*

Participants in Group 2, who interacted with a conversational agent featuring intermediate gaze-awareness capabilities (e.g., recognizing areas of interest within paintings), reported significantly higher **enjoyment** compared to Group 1 (basic gaze-awareness). This suggests that an agent capable of responding to more nuanced user behaviors enhances interaction enjoyment. However, enjoyment levels in Group 3 (advanced gaze pattern analysis) did not significantly exceed those in Group 2, indicating that while intermediate gaze-awareness improves enjoyment, more advanced capabilities might not yield a proportional increase in perceived enjoyment. This plateau effect highlights the importance of balancing technological complexity with user expectations and interac-

tion design. Despite differences in gaze-awareness levels, user **engagement** scores did not differ significantly across the groups. This suggests that factors other than gaze-awareness, such as the conversational agent's verbal responses or the quality of the VR environment, may play a more prominent role in sustaining user engagement.

The integration of real-time gaze data had a significant impact on **perceived personalization**. Group 3, which featured the most advanced gaze-awareness capabilities, was rated significantly higher in personalization compared to both Group 1 and Group 2. This finding underscores the importance of leveraging detailed gaze patterns to create more tailored and adaptive interactions. The ability of the agent in Group 3 to adapt to users' focus appears to have enhanced their perception of being understood and catered to, which is critical for fostering meaningful interactions in VR. During the interviews, participants were asked whether they felt the interaction was personalized. While some mentioned that gaze adaptation and question responses contributed to personalization, the agent did not evoke a strong sense of personalization. Participants felt more aware of the agent's actions rather than experiencing a deeper personal connection or customized content. The **sense of collaboration** followed a similar trend to perceived personalization. Participants in Group 3 reported significantly higher collaboration scores than those in Group 1, indicating that advanced gaze-awareness capabilities contribute to a stronger feeling of working together with the agent. This suggests that as conversational agents become more attuned to users' focus and behavior, they are perceived more as collaborative partners rather than mere informational tools.

Despite differences in functionality, participants' perceptions of the agent's **gaze awareness** did not vary significantly across groups. This suggests users may struggle to notice distinctions in gaze-aware behavior or assume a default level of awareness. Most participants believed the agent accurately identified the painting they were viewing. On the second platform, some noticed that the agent recognized specific areas of interest. However, some also felt that their time spent on specific areas was considered. Regarding the third platform, participants rarely perceived the agent as considering their gaze patterns. While many felt the agent identified the painting correctly, they believed it struggled to detect specific areas or behaviors. This may be due to the nature of the task—question answering typically does not require gaze input, which could explain the lack of perceived gaze awareness. Participants were more likely to recognize gaze-aware behavior during agent-initiated prompts, though this awareness typically surfaced only in post-session interviews or questionnaires, rather than during the interaction itself. Finally, it is important to note that the limited number of participants per platform ( $n = 17$ ) was one of the study's limitations.

**Conclusion** Integrating detailed gaze-awareness into a conversational agent within VR art exhibitions significantly enhances perceived personalization and collaboration. However, increasing the sophistication of gaze analysis does not notably influence enjoyment, engagement, or perceived gaze awareness, as participants found intermediate gaze-awareness (i.e., recognizing specific areas of interest) sufficiently enjoyable. While gaze-awareness capabilities show clear promise, their effectiveness depends on thoughtful integration with other design elements.

To advance hybrid intelligent systems, future designs should support shared agency—where users see the agent as a responsive partner rather than a passive guide. Prioritizing user autonomy, reducing cognitive load, and improving the clarity of agent behavior can foster richer, more adaptive interactions in cultural heritage settings.

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